Q1

i **Prove that $R(k_3, k_5) = 14$**

In order to prove that $R(k_3, k_5) = R(3,5) = 14$, we will show that

(i) $R(3,5) \leq 14$, and

(ii) $R(3,5) > 13$.

Case (i): $R(3,5) \leq 14$.

The proof of theorem 3.7 and corollary 3.8 provide an upper bound for Ramsey numbers such that $R(s,t) \leq R(s,t-1) + R(s-1,t)$ for $s, t \geq 2$. For $R(3,5)$ we have that

$$R(3,5) \leq R(2,5) + R(3,4).$$

For $R(2,5)$ we have by proposition 3.6 and symmetry of Ramsey numbers that $R(2,t) = t$, so $R(2,5) = 5$. We know from proposition 3.9 that $R(3,4) = 9$ (see proof in notes that draws on proposition 3.6, corollary 3.8, and theorem 3.1).

So we have the upper bound

$$R(3,5) \leq R(2,5) + R(3,4)$$
$$= 5 + 9 = 14.$$

Since $R(2,5) = 5$ and $R(3,4) = 9$ are both odd, we have by corollary 3.8 that the inequality need not be strict, so $R(3,5) \leq 14$.

Case (ii): $R(3,4) > 13$.

Similar to the proof of proposition 3.9, it is equivalent to state that what remains to be shown is that $R(3,4) \neq 13$.

For the existence of a graph on 13 vertices that contains no K_3 and no independent set of size 5, we turn to a known result in Ramsey theory, namely the Greenwood-Gleason graph. In this graph, we consider the vertices labelled as integers modulo 13, $\{0, 1, \ldots, 13\}$. Two vertices are adjacent if $x - y \equiv \pm 1 \pmod{13}$ or $x - y \equiv \pm 5 \pmod{13}$. It can be verified that this graph is triangle-free, and that its largest independent set is size 4 (see figure 1 below).

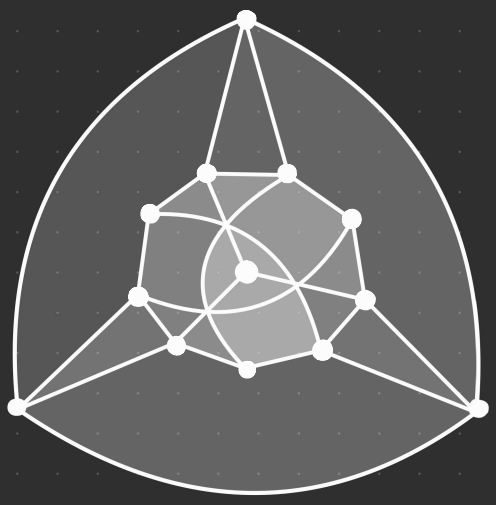


Figure 1: The Greenwood-Gleason graph G_{13}.

Since $R(3,5) \leq 14$ and $R(3,5) > 13$, we conclude that $R(3,5) = 14$.

i. **Prove that $R(P_3, K_3) = 5$**

To prove that $R(P_3, K_3) = 5$, we need to show:

(i) $R(P_3, K_3) \leq 5$

(ii) $R(P_3, K_3) > 4$

Case (i): $R(P_3, K_3) \leq 5$

Let K_5 be a complete graph on 5 vertices, and let its edges be coloured blue and red. We seek to show that there must be a red P_3 or blue K_3 subgraph.

Suppose there is no red P_3. A graph with no P_3 must have maximum degree at most one (if a vertex v has two neighbours u, w, then $\{u, v, w\}$ is a red P_3). A graph on 5 vertices where every vertex can have degree at most 1 can have at most $\lfloor \frac{5}{2} \rfloor = 2$ edges (since each of the edges requires 2 vertices). So the number of edges in K_5 is $\binom{5}{2} = 10$, while the red subgraph of K_5 without P_3 has 2 edges. This means that the blue subgraph contains $10 - 2 = 8$ edges. By Mantel's theorem (theorem 3.18), we have that a graph on $n \geq 3$ vertices with more than $\lfloor \frac{n^2}{4} \rfloor$ edges contains a K_3. For $n = 5$, $\lfloor \frac{n^2}{4} \rfloor = \lfloor \frac{25}{4} \rfloor = 6$. Since the blue graph has 8 edges, and $8 > 6$, it must contain a triangle. Thus, if there is no P_3, there must be a K_3. Thus $R(P_3, K_3) \leq 5$.

Case (ii): $R(P_3, K_3) > 4$

To show $R(P_3, K_3) > 4$, we must find a red-blue colouring such that there does not exist a red P_3 or blue K_3. Let the vertices of K_4 be labelled $\{1, 2, 3, 4\}$. Colour the edges $(1,2)$ and $(3,4)$ red. Colour all other edges blue. The blue edges are then $(1,3), (1,4), (2,3), (2,4)$ as seen in figure 2 below.

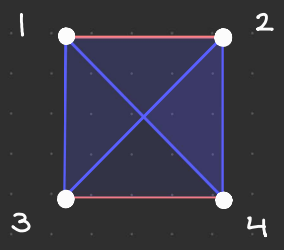


Figure 2: Red-blue-colouring of K_4

Here, the red graph consists of 2 disjoint edges, where no vertex has degree 2 or more, so by the reasoning outined in case (i), there is no red P_3.

The blue graph consists of a cycle of length 4 which is bipartite, which does not contain any triangles. So there is no blue K_3.

Since we have found a 2-colouring of K_4 that avoids both P_3 and K_3, it follows that $R(P_3, K_3) > 4$.

Combining both cases, we have that $R(P_3, K_3) \leq 5$, and $R(P_3, K_3) > 4$, therefore $R(P_3, K_3) = 5$.

Q2 Prove using induction that every graph on $\binom{k+\ell}{k}$ vertices contains either a complete subgraph on $k+1$ vertices or an independent set containing $\ell+1$ vertices.

Let $P(k,\ell)$ be the assertion that $R(k+1,\ell+1) \leq \binom{k+\ell}{k}$. We proceed to prove $P(k,\ell)$ for all $k,\ell \geq 0$ by double induction.

For the base cases we consider when $k=\ell=0$.

If $k=0$, we must show $P(0,\ell): R(1,\ell+1) \leq \binom{\ell}{0} = 1$.

From proposition 3.5, we have that $R(s,1)=1$ for $s \geq 1$.

By symmetry, $R(1,t)=1$ for any $t \geq 1$. Since $\ell \geq 0$, $\ell+1 \geq 1$, so $R(1,\ell+1)=1$. Thus $1 \leq 1$, so $P(0,\ell)$ holds for all $\ell \geq 0$.

If $\ell=0$, we must show $P(k,0): R(k+1,1) \leq \binom{k}{k} = 1$.

Again, by proposition 3.5 we know that $R(s,1)=1$.

Since $k \geq 0$, $k+1 \geq 1$, so $R(k+1,1)=1$. Thus $1 \leq 1$, so $P(k,0)$ holds for all $k \geq 0$.

Now assume that $P(k-1,\ell)$ and $P(k,\ell-1)$ hold for $k,\ell \geq 1$. We have that

$$P(k-1,\ell) \Rightarrow R(k,\ell+1) \leq \binom{(k-1)+\ell}{k-1} = \binom{k+\ell-1}{k-1}, \quad \text{IH1}$$

and

$$P(k,\ell-1) \Rightarrow R(k+1,\ell) \leq \binom{k+(\ell-1)}{k} = \binom{k+\ell-1}{k}. \quad \text{IH2}$$

We aim to show $P(k,\ell): R(k+1,\ell+1) \leq \binom{k+\ell}{k}$.

Corollary 3.8 gives us that for $s,t \geq 2$,

$$R(s,t) \leq R(s-1,t) + R(s,t-1).$$

Let $s = k+1$ and $t = \ell+1$. Since $k,\ell \geq 1$, we have $s,t \geq 2$. So $R(k+1,\ell+1) \leq R(k,\ell+1) + R(k+1,\ell)$.

Applying IH1 and IH2, we find

$$R(k+1,\ell+1) \leq \binom{k+\ell-1}{k-1} + \binom{k+\ell-1}{k}.$$

By Pascal's identity, $\binom{n'}{j-1} + \binom{n'}{j} = \binom{n'+1}{j}$. Setting $n' = k+\ell-1$ and $j = k$ we derive

$$\binom{k+\ell-1}{k-1} + \binom{k+\ell-1}{k} = \binom{k+\ell}{k}.$$

Thus, $R(k+1,\ell+1) \leq \binom{k+\ell}{k}$.

By the principle of mathematical induction, $P(k,\ell)$ holds for all $k,\ell \geq 0$. This means that any graph on $\binom{k+\ell}{k}$ vertices contains either a K_{k+1} subgraph or an independend set of $(\ell+1)$ vertices.

Q3 Show that a graph on n vertices and m edges has at least $\frac{m}{3n}(4m-n^2)$ triangles.

Let $G=(V,E)$ be a graph on $n=|V|$ vertices and $m=|E|$ edges. Let t denote the number of triangles in G. For $u,v \in V$ and $\{u,v\} \in E$, let t_{uv} be the number of triangles containing the edge $\{u,v\}$. Summing over all edges, every triangle is counted 3 times (once for each edge), so

$$\sum_{\{u,v\} \in E} t_{uv} = 3t.$$

A triangle containing $\{u,v\}$ arises precisely when some vertex w is a common neighbour of both u and v. Thus,

$$t_{uv} = |N(u) \cap N(v)|,$$

where $N(u)$ denotes the neighbours of u and similarly for v. To bound this from below, observe that for any u,v, by the principal of inclusion-exclusion,

$$|N(u) \cap N(v)| = |N(u)| + |N(v)| - |N(u) \cup N(v)|.$$

Now, clearly $|N(u)| = d(u)$, $|N(v)| = d(u)$, and $|N(u) \cup N(v)| \leq n$ (since each neighbour is a vertex other than possibly u and v). Therefore,

$$|N(u) \cap N(v)| \geq d(u) + d(v) - n.$$

Summing over all edges,

$$3t = \sum_{\{u,v\} \in E} t_{uv} \geq \sum_{\{u,v\} \in E} (d(u)+d(v)-n) = \sum_{\{u,v\} \in E} (d(u)+d(v)) - mn$$

Now, observe that in $\sum_{\{u,v\}\in E}(d(u)+d(v))$, each degree $d(x)$ is summed once for each edge incident to x, that is, $d(x)$ times, so

$$\sum_{\{u,v\}\in E}(d(u)+d(v)) = \sum_{x\in V} d(x)^2.$$

Therefore,

$$3t \geq \sum_{x\in V} d(x)^2 - mn.$$

Next, apply Cauchy-Schwarz inequality:

$$\left(\sum_{x\in V} d(x)\right)^2 \leq n \sum_{x\in V} d(x)^2.$$

Since $\sum_{x\in V} d(x) = 2m$ (by the Handshaking lemma), this gives

$$(2m)^2 \leq n \sum_{x\in V} d(x)^2 \Rightarrow \sum_{x\in V} d(x)^2 \geq \frac{4m^2}{n}.$$

So,

$$3t \geq \frac{4m^2}{n} - mn,$$

$$t \geq \frac{1}{3}\left(\frac{4m^2}{n} - mn\right) = \frac{4m^2 - mn^2}{3n} = \frac{m}{3n}\left(4m - n^2\right),$$

as required.

Q4 Using combinatorial counting, show that the Turán graph $T(n,r)$ has at most $(1-\frac{1}{r})\frac{n^2}{2}$ edges.

Consider the Turán graph $T(n,r)$ whose vertex set V, with $|V| = n$, is partitioned into r independent sets $V_1, V_2, \ldots, V_r$. Let $n_i = |V_i|$ for $i = 1, \ldots, r$. The cardinalities n_i are chosen to be as equal as possible, such that they differ from each other by at most 1. Specifically, if $n = qr + s$, where $q = \left\lfloor \frac{n}{r} \right\rfloor$ and $0 \leq s < r$, then s of these sets have size $q+1$ and the remaining $r-s$ sets have size q. It follows that $\sum_{i=1}^{r} n_i = s(q+1) + (r-s)q = n$.

The number of edges in $T(n,r)$, denoted $t(n,r)$, is the total number of pairs less the number of pairs within each partition set (as these edges are absent). So we have

$$t(n,r) = \binom{n}{2} - \sum_{i=1}^{r} \binom{n_i}{2}.$$

Using the identity $\binom{k}{2} = \frac{k!}{2!(k-2)!} = \frac{k(k-1)}{2}$, we can write

$$t(n,r) = \frac{n(n-1)}{2} - \sum_{i=1}^{r} \frac{n_i(n_i-1)}{2}$$

$$= \frac{1}{2}\left(n(n-1) - \sum_{i=1}^{r} (n_i^2 - n_i) \right).$$

This simplifies to

$$t(n,r) = \frac{1}{2}\left(n^2 - n - \sum_{i=1}^{r} n_i^2 + \sum_{i=1}^{r} n_i \right).$$

Since $\sum_{i=1}^{r} n_i = n$, this further simplifies to

$$t(n,r) = \frac{1}{2}\left(n^2 - \sum_{i=1}^{r} n_i^2\right).$$

To evaluate $\sum_{i=1}^{r} n_i^2$, we use the distribution of partition sizes: s sets of size $q+1$ and $r-s$ sets of size q. So

$$\begin{aligned}\sum_{i=1}^{r} n_i^2 &= s(q+1)^2 + (r-s)q^2 \\ &= s(q^2+2q+1) + (r-s)q^2 \\ &= sq^2 + 2sq + s + rq^2 - sq^2 \\ &= rq^2 + 2sq + s.\end{aligned}$$

Substituting $q = \frac{(n-s)}{r}$ (from $n = qr + s$), we obtain

$$\begin{aligned}\sum_{i=1}^{r} n_i^2 &= r\left(\frac{n-s}{r}\right)^2 + 2s\left(\frac{n-s}{r}\right) + s \\ &= \frac{(n-s)^2}{r} + \frac{2s(n-s)}{r} + \frac{rs}{r}.\end{aligned}$$

Combining terms over a common denominator

$$\begin{aligned}\sum_{i=1}^{r} n_i^2 &= \frac{n^2 - 2ns + s^2 + 2sn - 2s^2 + rs}{r} \\ &= \frac{n^2 - s^2 + rs}{r} \\ &= \frac{n^2 + s(r-s)}{r}.\end{aligned}$$

Now, substituting this expression for $\sum_{i=1}^{r} n_i^2$ into the formula for $t(n,r)$ we get that

$$t(n,r) = \frac{1}{2}\left(n^2 - \frac{n^2 + s(r-s)}{r}\right)$$

$$= \frac{1}{2}\left(\frac{rn^2 - n^2 - s(r-s)}{r}\right).$$

This can be rewritten as

$$t(n,r) = \frac{(r-1)n^2 - s(r-s)}{2r}$$

$$= \frac{r-1}{r} \cdot \frac{n^2}{2} - \frac{s(r-s)}{2r}.$$

Factoring $\frac{(r-1)}{r}$ as $1 - \frac{1}{r}$, we arrive at

$$t(n,r) = \left(1 - \frac{1}{r}\right) \cdot \frac{n^2}{2} - \frac{s(r-s)}{2r}.$$

To prove the inequality $t(n,r) \leq \left(1 - \frac{1}{r}\right)\frac{n^2}{2}$, we examine the term $\frac{s(r-s)}{2r}$. By definition of s as the remainder when n is divided by r, we have $0 \leq s < r$.

If $s = 0$, then $\frac{s(r-s)}{2r} = 0$.

If $s > 0$, then since $s < r$, it implies $r - s > 0$. Also, $r \geq 1$, since it is the number of partitions.

Thus, $s \geq 0$, $r - s > 0$ (unless $s = 0$, in which case the product is 0), and $2r > 0$. Consequently, the product $s(r-s) \geq 0$, and therefore $\frac{s(r-s)}{2r} \geq 0$. Since $t(n,r) = \left(1 - \frac{1}{r}\right)\frac{n^2}{2} - \frac{s(r-s)}{2r}$, and $\frac{s(r-s)}{2r} \geq 0$, if follows directly that $t(n,r) \leq (1 - 1/r)\, n^2/2$.

Q5 Let G be a graph on n vertices and m edges.

i Prove that if $n \geq 3$ and $m \geq \binom{n-1}{2} + 2$, then G contains a Hamiltonian cycle. Hint: Use Ore's theorem.

We aim to prove that a (simple) graph G with n vertices and m edges, where $n \geq 3$ and $m \geq \binom{n-1}{2} + 2$, contains a Hamiltonian cycle. First we establish the connectivity of G. A graph with n vertices can have at most $\binom{n-1}{2}$ edges if it is disconnected (consider K_{n-1} and K_1). Since the given graph G has $m \geq \binom{n-1}{2} + 2 > \binom{n-1}{2}$ edges, G cannot be disconnected and therefore must be connected.

Now, we verify the degree sum condition for any pair of distinct, non-adjacent vertices $u, v \in V$. Let u and v be such a pair. The set of remaining $n-2$ vertices is denoted by $V' = V \setminus \{u, v\}$. The edges of G can be categorised to those entirely within V', those connecting u to V', and those connecting v to V'. Since u and v are non-adjacent, all $d(u)$ edges incident to u connect to vertices in V', and similarly, all $d(v)$ edges incident to v connect to vertices in V'. The number of edges with both endpoints in V', say $m_{V'}$, is at most $\binom{n-2}{2}$, the number of edges in a complete graph on $n-2$ vertices.

The total number of edges m in G is

$$m = m_{V\setminus} + d(u) + d(v).$$

From this, we have that

$$m \leq \binom{n-2}{2} + d(u) + d(v).$$

We are given $m \geq \binom{n-1}{2} + 2$. Combining these inequalities we obtain

$$\binom{n-1}{2} + 2 \leq d(u) + d(v) + \binom{n-2}{2}.$$

Rearranging for the sum of degrees gives

$$d(u) + d(v) \geq \binom{n-1}{2} - \binom{n-2}{2} + 2.$$

We evaluate the difference of the binomial coefficients:

$$\begin{aligned}\binom{n-1}{2} - \binom{n-2}{2} &= \frac{(n-1)(n-2)}{2} - \frac{(n-2)(n-3)}{2} \\ &= \frac{n-2}{2}\left((n-1)-(n-3)\right) \\ &= \frac{n-2}{2} \cdot 2 \quad = n-2.\end{aligned}$$

Substituting this back into the inequality for $d(u) + d(v)$ we get that

$$d(u) + d(v) \geq (n-2) + 2 = n. \quad \text{(Ore's theorem)}.$$

This holds for any pair of non-adjacent vertices u and v. Since G is a simple graph with $n \geq 3$ vertices, is connected, and for every pair of non-adjacent vertices u, v, we have $d(u) + d(v) \geq n$, all conditions of Ore's theorem are satisfied, and therefore we conclude that G contains a Hamiltonian cycle.

(ii) Hence determine the smallest value of m for which G contains a Hamiltonian path.

To determine the smallest value of m for which G contains a Hamiltonian path, we need to find a value m_0 such that

(i) Any graph G with n vertices and $m \geq m_0$ edges has a Hamiltonian path, and

(ii) There exists a graph G' with n vertices and $m_0 - 1$ edges that does not have a Hamiltonian path.

Let this smallest value be $m_{HP}(n)$.

Consider the graph $G_0 = K_{n-1} \cup K_1$. The number of vertices in G_0 is $(n-1) + 1 = n$. The number of edges in K_{n-1} is $\binom{n-1}{2}$, while the isolated vertex contains no edges, so G_0 has $m = \binom{n-1}{2}$ edges.

A hamiltonian path must traverse every vertex. Since G_0 has an isolated vertex (for $n \geq 2$), no path can visit both this isolated vertex and any vertex in the K_{n-1} component. Thus G_0 does not contain a Hamiltonian path. This implies that $m_{HP}(n)$ must be strictly greater than $\binom{n-1}{2}$, so we have that $m_{HP}(n) \geq \binom{n-1}{2} + 1$.

We now seek to show that if a graph G on $n \geq 3$ vertices has $m \geq \binom{n-1}{2} + 1$ edges, it satisfies the

conditions of a known analogue to Ore's theorem (which is a generalisation of Dirac's theorem), for Hamiltonian paths. This theorem states that if a simple graph on $n \geq 2$ vertices is connected and for every pair of non-adjacent vertices $u, v \in V$, we have $d(u) + d(v) \geq n-1$, then G has a Hamiltonian path.

First, we establish connectivity. As shown in Q5 part (i), a graph with $\binom{n-1}{2}$ edges can be disconnected. A graph on n vertices and $m > \binom{n-1}{2}$ must be connected. Since $m \geq \binom{n-1}{2} + 1 > \binom{n-1}{2}$, G is connected.

Next, we show the degree sum condition. Let u and v be any two distinct, non-adjacent vertices in G. Let $V' = V \setminus \{u, v\}$ be the set of the other $(n-2)$ vertices. The edges of G can be divided into:

(a) Edges within the induced subgraph $G[V']$. Let this number be $m_{V'}$. The maximum possible value for $m_{V'}$ is $\binom{n-2}{2}$ (the number of edges in K_{n-2}).

(b) Edges connecting u to vertices in V'. Since u and v are not adjacent, all $d(u)$ edges incident to u connect to V'. So there are $d(u)$ such edges.

© Edges connecting v to vertices in V'. Similarly, there are $d(v)$ such edges.

The total number of edges m in G is

$$m = m_{V'} + d(v) + d(u).$$

Since $m_{V'} \leq \binom{n-2}{2}$, we have $m \leq \binom{n-2}{2} + d(u) + d(v)$.
We are given $m \geq \binom{n-1}{2} + 1$. Combining these,

$$\binom{n-1}{2} + 1 \leq \binom{n-2}{2} + d(u) + d(v).$$

Rearranging for the sum of degrees:

$$d(u) + d(v) \leq \binom{n-1}{2} - \binom{n-2}{2} + 1.$$

We calculate the difference of the binomial coefficients:

$$\begin{aligned}\binom{n-1}{2} - \binom{n-2}{2} &= \frac{(n-1)(n-2)}{2} - \frac{(n-2)(n-3)}{2} \\ &= \frac{n-2}{2}\Big((n-1) - (n-3)\Big) \\ &= \frac{n-2}{2} \cdot 2 \\ &= n-2.\end{aligned}$$

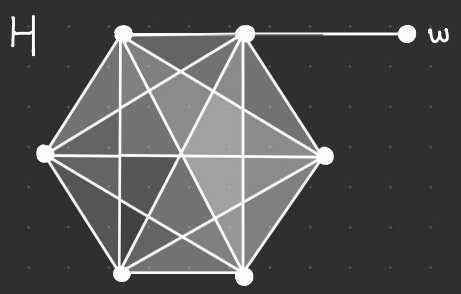
Substituting this back in, we get

$$d(u) + d(v) \geq (n-2) + 1 = n-1.$$

Thus, for any non-adjacent pair of vertices u and v in a graph G on $n \geq 3$ vertices and $m \geq \binom{n-1}{2} + 1$ edges, we satisfy the degree-sum condition derived from Ore's theorem, and conclude that G contains a Hamiltonian path.

iii Give examples of extremal graphs for parts (i) and (ii).

(i) Extremal graph for Hamiltonian cycle: $\binom{n-1}{2} + 1$ edges



$K_6 \subset H$

Consider H which a graph on 7 vertices consisting of a K_6 subgraph with one additional vertex w_2 and incident edge to w_1 in the K_6 subgraph.

The number of edges of H is

$$m_H = \underbrace{\binom{n-1}{2}}_{\text{from } K_6} + \underbrace{1}_{\{w_1, w_2\}} = \binom{6}{2} + 1 = 15.$$

For any graph with $n \geq 3$ vertices, to hold a Hamiltonian cycle, every vertex must have degree at least 2 (if a vertex is part of a cycle, it must have two distinct edges in the cycle incident to it). Since $\deg(w_2) = 1$ and $n \geq 3$, H cannot contain any cycle. By the results of part (i), H is extremal for a Hamiltonian cycle.